

Problem

The terminal voltage of a 2-H inductor is $v = 10(1 - t)$ V. Find the current flowing through it at $t = 4$ s and the energy stored in it at $t = 4$ s. Assume $i(0) = 2$ A.

Step-by-step solution

Step 1 of 2

Given that,

$$L = 2 \text{ H}$$

$$v = 10(1 - t) \text{ V}$$

Since,

$$i = \frac{1}{L} \int_0^t v(t) dt + i(0)$$

$$i = \frac{1}{2} \int_0^4 10(1 - t) dt + 2 \quad (\because i(0) = 2 \text{ A})$$

$$i = \frac{1}{2} \left[\int_0^4 10 dt - \int_0^4 10t dt \right] + 2$$

$$i = \frac{1}{2} \left[\left[10t \right]_0^4 - \left[10 \frac{t^2}{2} \right]_0^4 \right] + 2$$

$$i = \frac{1}{2} \left[10(4 - 0) - 10 \frac{(4)^2}{2} - 0 \right] + 2$$

$$i = \frac{1}{2} [40 - 80] + 2$$

$$\therefore \boxed{i = -18 \text{ A}}$$

Step 2 of 2

The energy stored is,

$$w = W \Big|_4 - W \Big|_0$$

$$w = \frac{1}{2} L [i(5)]^2 - \frac{1}{2} L [i(0)]^2$$

$$w = \left[\frac{1}{2} \times 2 \times (-18)^2 \right] - \left[\frac{1}{2} \times 2 \times (2)^2 \right]$$

$$\therefore \boxed{w = 320 \text{ J}}$$